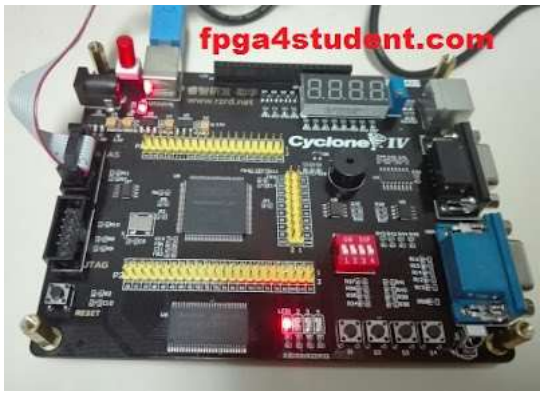
Altera FPGA boards for beginners

Last time, I presented my four recommended and affordable *Xilinx FPGA boards* for beginners. The recommended Xilinx FPGA boards offer good enough number of IO devices and supporting circuits for student projects or practice, and more importantly, the FPGA boards are affordable for beginners or students.

In this FPGA post, I will present another four recommended and affordable *Altera FPGA boards* for beginners. These Altera FPGA boards are also very cheap and well-designed for students or beginners.

1. Cyclone IV Altera FPGA Board from Aliexpress (33.69USD)



Following are the good points from the Altera Cyclone IV FPGA Board to make this FPGA board recommended and affordable for beginners:

- **On board Altera FPGA Cyclone IV: EP4CE6E22C8N**
- **Only 33.69USD: affordable Altera FPGA Board for beginners**
- **4 single LEDs, 4 buttons, 4 Switches, 4 7-segment LEDs, other 1x20 IOs for LCD**
- **On board VGA port, RS232 port for UART communication**
- **On board PS/2, I2C, infrared receiver, 64Mb SDRAM**
- **Supports FPGA configuration via JTAG/AS programming mode (USB-Blaster cable provided)**
- **Good feedback and reviewed by hundreds of students**

Obviously, the affordable Altera FPGA board provide many IO devices and enough peripherals for beginners to play and practice. The FPGA board supplier also includes the USB-Blaster programming cables in the FPGA board package. With the price of only 33.69USD, this FPGA board is a very good choice for beginners . That's why the FPGA board has attracted hundreds of students at the moment.

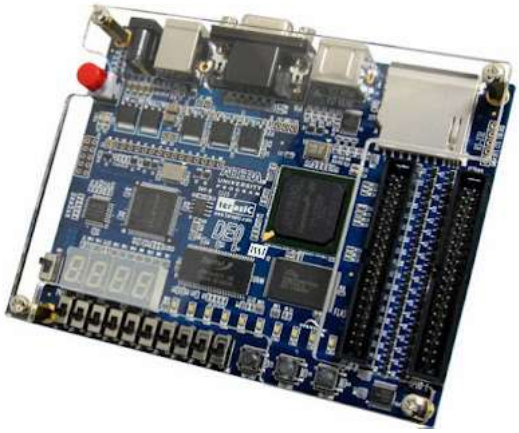
2. ZEOWAA Cyclone IV Altera FPGA Board from Aliexpress 34.9USD

Below are the reasons why this Altera FPGA board is a good choice and recommended for beginners:

- On board Altera FPGA Cyclone IV: EP4CE10E22C8N
- Only 34.9USD: an affordable Altera FPGA board for beginners, unfortunately, it is not available in stock at the moment.
- 4 keys, 12 single LEDs, 8 7-segment LEDs, other 20 IOs for LCD
- On board PS/2, VGA, RS232 port for UART communications
- Offer sample code and programming file to test the Altera FPGA board
- The FPGA board also supports FPGA configuration via JTAG/AS programming mode (USB-Blaster cable provided)

This Altera FPGA board provides a higher number of peripherals and IOs for beginners to practice FPGA projects. The FPGA board supplier also gives sample Verilog/ VHDL code and .sof programming file for testing the FPGA board.

3. Altera FPGA Board DE0 From Terasic 81USD

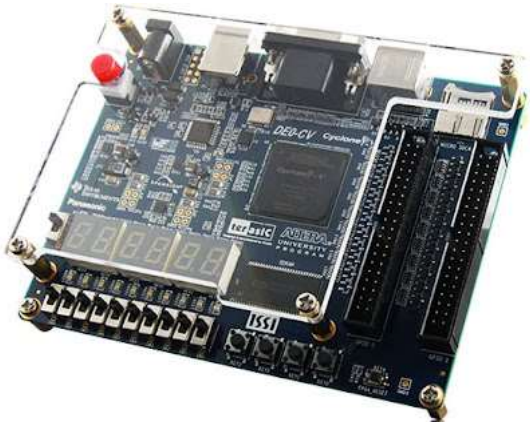


The DE0 Altera FPGA Board is well designed and suited for students projects in university. The good features from the Altera FPGA board to make it recommended and affordable for beginners are as follows:

- **Altera FPGA Cyclone III 3C16**
- **81$ affordable Altera FPGA Board (academic pricing for students), 119$ for normal price**
- **10 single green LEDs, 10 slide switches, 3 buttons, 4 7-segment LEDs, 40-pin expansion headers**
- **VGA port, RS232 interface, PS/2 port for keyboard/ mice**
- USB-Blaster onboard circuit for FPGA programming

This Altera FPGA board gives beginners many onboard IO devices and peripherals for practicing FPGA projects. This DE0 Altera FPGA board is also provided by Terasis, a very reliable FPGA board supplier, so beginners can get their full support on the development tool and documentation together with sample code and example FPGA projects.

4. DE0-CV Altera FPGA Board 99USD



This DE0-CV Altera FPGA Board is also a reliable Altera FPGA board from Terasis as the DE0-Nano FPGA board. With a bit higher price, this Altera FPGA board provides even better features for beginners as follows:

- **Altera FPGA Cyclone V: 5CEBA4F23C7N**
- **99$ affordable Altera FPGA Board (academic pricing for students), 150$ for normal price**
- **10 single LEDs, 10 switches, 4 buttons, 6 7-segment LEDs**
- **Micro SD card, 62MB SDRAM, VGA port, PS/2 port**
- Other expansions GPIOs for other user defined interface
- USB-Blaster onboard circuit for programming (JTAG/AS mode)

This DE0-CV Altera FPGA is quite similar to the Basys-3 Xilinx FPGA Board. The FPGA board provides perfect IO devices and other supporting circuits for academic projects. Although the price for the FPGA board is a bit higher than the other 3 Altera FPGA boards, this Altera FPGA board offers more IO and peripherals devices so that beginners can practice with more options of verifications, and for sure, the FPGA board will be more fun to play with. Thus, this DE0-CV Altera FPGA board is a good and reliable choice for beginners.

Above are my four recommended and affordable Altera FPGA boards for beginners. These four Altera FPGA boards are not only affordable for beginners but also provide enough number of IOs and other peripherals for students practice.

Recommended and affordable Xilinx FPGA boards for beginners or students
Verilog Projects for beginners or students
VHDL projects for beginners students
What is FPGA Programming? FPGA vs Software Programming

Recommended FPGA projects for beginners:

1. What is an FPGA? How VHDL/Verilog works on FPGA
2. How to load a text file into FPGA using VHDL/ Verilog
3. 8-bit Microcontroller Design on FPGA using VHDL
4. Matrix Multiplication on Xilinx FPGA
5. Digital alarm clock on FPGA
6. Cryptographic Coprocessor Design on FPGA
7. 16-bit single-cycle MIPS processor design on FPGA
8. Microcontroller implementation on FPGA
9. Car Parking System on FPGA
10. Image processing on FPGA using Verilog HDL
11. Traffic Light Controller on FPGA
12. Alarm Clock on FPGA
13. Comparator design on FPGA using Verilog
14. 16-bit RISC Processor design on FPGA
15. Pipelined MIPS Processor Design on FPGA
16. Basys 3 FPGA OV7670 Camera

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